

Multi-weight genus expansion and cross-channel corrections

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Abstract

For a modular Koszul chiral algebra whose strong generators have a single conformal weight, the genus- g free energy satisfies the scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ at all genera (UNIFORM-WEIGHT). This paper proves that the scalar formula fails for multi-weight algebras at genus $g \geq 2$ and computes the failure explicitly. The full genus expansion decomposes as $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$ (ALL-WEIGHT), where $\delta F_g^{\text{cross}}$ is a graph sum over mixed-channel boundary strata of $\overline{\mathcal{M}}_{g,0}$. The cross-channel correction vanishes identically at genus 1, on the uniform-weight locus, and for all free-field algebras. For $\mathcal{W}_3$ we compute $\delta F_2 = (c+204)/(16c)$ over the 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$ and δF_3 over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$; the cross-channel correction is already comparable to the scalar part at genus 3. A universal closed-form N -formula for $\delta F_2(\mathcal{W}_N)$ is proved, with the Virasoro algebra ($N = 2$) as the unique point of vanishing. The scalar and cross-channel sectors have distinct resurgent structures: the scalar instanton action is the universal $A_{\text{scal}} = (2\pi)^2$, while the cross-channel growth exceeds it at large central charge. The cross-channel dominance is the quantitative expression of E_1 primacy: the ordered bar complex $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ resolves all channels simultaneously, and the cross-channel correction measures the information lost by the coinvariant projection $\text{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\text{mod}}$.

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I Introduction

1.1 The scalar formula and its failure

The modular Koszul programme associates to each chiral algebra $\mathcal{A}$ a modular characteristic $\kappa(\mathcal{A})$ and a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$. The genus- g free energy extracts the weight- g component of the MC element through the stable-graph expansion on $\overline{\mathcal{M}}_{g,n}$. When $\mathcal{A}$ has strong generators of a single conformal weight (the *uniform-weight* case: Heisenberg, Virasoro, affine Kac–Moody at any single current weight), the genus expansion collapses to the scalar formula

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} \quad \text{for all } g \geq 1 \quad (\text{UNIFORM-WEIGHT}),$$

where $\lambda_g^{\text{FP}} = (2^{2g-1} - 1) |B_{2g}| / (2^{2g-1} \cdot (2g)!)$ is the Faber–Pandharipande intersection number $\int_{\overline{\mathcal{M}}_g} \lambda_g$ ($\lambda_1^{\text{FP}} = 1/24$, $\lambda_2^{\text{FP}} = 7/5760$, $\lambda_3^{\text{FP}} = 31/967680$).

The scalar formula encodes a profound universality: the genus- g free energy is a single number $\kappa(\mathcal{A})$ multiplied by a universal topological constant λ_g^{FP} . All information about the OPE, the representation category, and the symmetry algebra is compressed into the single invariant κ .

The question forced by the structure is whether this universality survives the passage to multiple generators at distinct conformal weights. The $\mathcal{W}_3$ algebra has two generators: T at conformal weight 2 and W at conformal weight 3. Its modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$, and the scalar formula predicts $F_2(\mathcal{W}_3) = (5c/6) \cdot (7/5760) = 7c/6912$. The actual genus-2 free energy is

$$F_2(\mathcal{W}_3) = \underbrace{\frac{7c}{6912}}_{\kappa \cdot \lambda_2^{\text{FP}} \text{ (UNIFORM-WEIGHT)}} + \underbrace{\frac{c + 204}{16c}}_{\delta F_2^{\text{cross}} \text{ (ALL-WEIGHT)}}.$$

The scalar formula fails. The correction $\delta F_2^{\text{cross}} = (c + 204)/(16c)$ is nonzero for all finite central charge $c > 0$ and resolves the multi-generator universality problem in the negative.

1.2 Scope and main results

This paper develops the theory of the cross-channel correction $\delta F_g^{\text{cross}}$ from first principles and computes it in the simplest non-trivial cases. The main results are:

- (1) The decomposition $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}} \text{ (ALL-WEIGHT)}$ holds for every modular Koszul chiral algebra at every genus $g \geq 1$; the diagonal part is universal and the cross-channel part is the sole obstruction to the scalar formula (Theorem 2.2).

- (2) The cross-channel correction vanishes identically at genus 1 (single-edge argument), on the uniform-weight locus (channel symmetry), and for all free-field algebras (triple redundancy: shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation).
- (3) For $\mathcal{W}_3$ at genus 2: $\partial F_2 = (c+204)/(16c)$, computed graph by graph over all 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$, of which 3 contribute nonzero cross-channel terms (Computation 3.1).
- (4) For $\mathcal{W}_3$ at genus 3: $\partial F_3 = (5c^3 + 3792c^2 + 1,149,120c + 217,071,360)/(138,240c^2)$, computed over all 42 stable graphs, of which 34 contribute (Computation 4.1). The cross-channel correction is comparable to the scalar part: the ratio $\partial F_3/(\kappa \cdot \lambda_3^{\text{FP}}) \rightarrow 42/31 \approx 1.35$ as $c \rightarrow \infty$.
- (5) A universal closed-form N -formula for the genus-2 gravitational cross-channel correction of $\mathcal{W}_N$ (Proposition 6.2), with $N = 2$ (Virasoro) as the unique vanishing point.
- (6) The cross-channel instanton hierarchy: the scalar sector has universal instanton action $\mathcal{A}_{\text{scal}} = (2\pi)^2$ (Gevrey-1), while the cross-channel growth at large c exceeds the scalar growth (Proposition 5.2).
- (7) The E_1 interpretation: the cross-channel correction measures the information in the ordered bar complex $B^{\text{ord}}(\mathcal{A})$ that is killed by the coinvariant projection to the symmetric bar $B^\Sigma(\mathcal{A})$ (§7).

1.3 Conventions

The bar complex is $B(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ with $\tilde{\mathcal{A}} = \ker(\varepsilon)$ the augmentation ideal, $|s^{-1}v| = |v| - 1$ (desuspension lowers degree), and deconcatenation coproduct. All grading is cohomological ($|d| = +1$). The ordered bar $B^{\text{ord}}(\mathcal{A})$ is the primitive E_1 -chiral coalgebra; the symmetric bar $B^\Sigma(\mathcal{A})$ is its Σ_n -coinvariant shadow. The modular characteristic $\kappa(\mathcal{A})$ is $\text{av}(r(z))$ at degree 2, where $r(z)$ is the classical r -matrix and $\text{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\text{mod}}$ is the (lossy) coinvariant projection. The Koszul scalar complementarity is $\varkappa(\mathcal{A}) := \kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$: $\varkappa = 0$ for Kac–Moody and free fields, $\varkappa = 13$ for Virasoro, $\varkappa = 250/3$ for $\mathcal{W}_3$. This scalar invariant is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^\dagger) = -c_{\text{ghost}}(\text{BRST})$ (values $K(\text{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\text{BP}) = 196$, $K(V_k(\mathfrak{g})) = 2 \dim \mathfrak{g}$); the two are related by $\varkappa = \rho_{\mathcal{A}} \cdot K$.

Every displayed formula involving F_g or obs_g carries one of the two tags: **(UNIFORM-WEIGHT)** when the formula holds under the hypothesis that all strong generators share a single conformal weight, or **(ALL-WEIGHT)** when the formula holds unconditionally (possibly with a cross-channel correction term).

2 The cross-channel correction

2.1 Multi-channel graph sum

Construction 2.1 (Multi-channel graph sum). Let $\mathcal{A}$ be a modular Koszul chiral algebra of rank r with strong generators $\varphi_1, \dots, \varphi_r$ at conformal weights $h_1, \dots, h_r$ and diagonal Zamolodchikov metric $\eta = \text{diag}(\eta_1, \dots, \eta_r)$. Fix a genus $g \geq 2$. For each stable graph $\Gamma \in \mathcal{G}_{g,0}$ with edge set $E(\Gamma)$ and vertex set $V(\Gamma)$:

Channel assignment. A function $\sigma: E(\Gamma) \rightarrow \{1, \dots, r\}$. There are $r^{|E(\Gamma)|}$ such assignments.

Multi-channel amplitude.

$$A_\Gamma(\sigma, \mathcal{A}) := \prod_{e \in E(\Gamma)} \eta^{\sigma(e)\sigma(e)} \cdot \prod_{v \in V(\Gamma)} V_{g(v), n(v)}(\sigma|_{H(v)}),$$

where $\eta^{ii} = 1/\eta_i$ is the inverse Zamolodchikov metric in channel i (the propagator; each edge is diagonal in the channel basis; weight 1 always), $V_{g(v),n(v)}(\sigma|_{H(v)})$ is the genus- $g(v)$ amplitude at vertex v with $n(v)$ half-edges, and $g(v)$ is the genus label of v .

Full graph sum.

$$F_g(\mathcal{A}) = \sum_{\Gamma \in \mathcal{G}_{g,0}} \frac{1}{|\text{Aut}(\Gamma)|} \sum_{\sigma: E(\Gamma) \rightarrow \{1, \dots, r\}} A_\Gamma(\sigma, \mathcal{A}).$$

Diagonal/cross decomposition. An assignment σ is *diagonal* if $\sigma(e) = i$ for all e (constant on all edges); otherwise it is *mixed*. The diagonal sum gives $F_g^{\text{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT: per-channel universality; each single-channel component has uniform weight). The mixed sum gives $\delta F_g^{\text{cross}}(\mathcal{A})$.

2.2 The decomposition theorem

Theorem 2.2 (Multi-weight genus expansion). *Let $\mathcal{A}$ be a modular Koszul chiral algebra with strong generators $\varphi_1, \dots, \varphi_r$ of conformal weights $h_1, \dots, h_r$, per-channel modular characteristics $\kappa_1, \dots, \kappa_r$, and total modular characteristic $\kappa(\mathcal{A}) = \sum_{i=1}^r \kappa_i$.*

(i) Per-channel universality. *The diagonal contribution to the genus- g free energy satisfies*

$$F_g^{\text{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} \quad \text{for all } g \geq 1 \quad (\text{UNIFORM-WEIGHT scalar lane}).$$

(ii) Cross-channel decomposition. *The full genus- g free energy decomposes as*

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}(\mathcal{A}) \quad \text{for all } g \geq 1 \quad (\text{ALL-WEIGHT}),$$

where $\delta F_g^{\text{cross}}(\mathcal{A})$ is the graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$ (Construction 2.1).

- (iii) Genus-1 universality. $\delta F_1^{\text{cross}} = 0$ for all $\mathcal{A}$: the unique genus-1 boundary graph (figure-eight) has a single edge, admitting only pure channel assignments.
- (iv) Uniform-weight universality. If $h_1 = h_2 = \dots = h_r$, then $\delta F_g^{\text{cross}} = 0$ for all $g \geq 1$.
- (v) R -matrix independence. The cross-channel correction $\delta F_g^{\text{cross}}$ is independent of the Givental R -matrix: genus-0 boundary vertices receive no R -contribution, and genus-1 vertices have vanishing R -correction for degree reasons.
- (vi) Free-field exactness. If $\mathcal{A}$ is a free-field algebra (purely quadratic genus-0 OPE: $m_k = 0$ for $k \geq 3$), then $\delta F_g^{\text{cross}}(\mathcal{A}) = 0$ for all $g \geq 1$ (ALL-WEIGHT free-field exact case).

Proof. (i) Each edge e of a stable graph Γ carries a single channel assignment $\sigma(e) \in \{1, \dots, r\}$. The diagonal graph sum restricts to constant assignments $\sigma(e) = i$ for all e . On such an assignment, the amplitude factors as $A_\Gamma(\sigma_i, \mathcal{A}) = A_\Gamma(\mathcal{A}_i)$, the single-channel amplitude for the rank-1 algebra $\mathcal{A}_i$ with curvature κ_i . Summing over all i and all graphs:

$$F_g^{\text{diag}}(\mathcal{A}) = \sum_{i=1}^r F_g(\mathcal{A}_i) = \sum_{i=1}^r \kappa_i \cdot \lambda_g^{\text{FP}} = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} \quad (\text{UNIFORM-WEIGHT}),$$

where the second equality uses single-channel universality: each single-channel component has a single generator and hence uniform weight.

(ii) The full graph sum runs over all channel assignments $\sigma: E(\Gamma) \rightarrow \{1, \dots, r\}$. The diagonal assignments give F_g^{diag} ; the mixed assignments give $\delta F_g^{\text{cross}}$.

(iii) At genus 1, the unique boundary graph is the figure-eight (one vertex of genus 0, one self-loop, $|E| = 1$). A single edge has only r channel assignments, each pure. No mixed-channel assignments exist on a single-edge graph.

(iv) When all conformal weights coincide, the channel permutation symmetry forces every mixed-channel amplitude to equal the corresponding diagonal amplitude; the difference vanishes.

(v) The genus-0 vertices receive transferred operations from the cyclic minimal model, which are independent of the Givental R -matrix by construction. Genus-1 vertices on $\overline{\mathcal{M}}_{1,1}$ have vanishing R -correction because the R -matrix acts in degree ≥ 2 on the state space and the genus-1 vertex amplitude requires only degree-0 and degree-1 data.

(vi) For free fields, three independent mechanisms each force $\delta F_g^{\text{cross}} = 0$:

Shadow-tower collapse. Free fields have $m_k = 0$ for $k \geq 3$, so every genus-0 vertex is binary. No higher-contact vertex couples generators of distinct conformal weights at a common vertex. Only diagonal assignments survive.

Off-diagonal metric elimination. The OPE pairing g^{ab} is block-diagonal with respect to conformal weight: $\varphi_a(z)\varphi_b(w) \sim 0$ for $h_a \neq h_b$ forces $g^{ab} = 0$. Each edge carries a propagator $P^{ab} = g^{ab} \cdot d \log E(z, w)$ (weight 1), so $P^{ab} = 0$ for cross-weight edges. Every mixed-channel graph has at least one such edge and therefore vanishes.

Ghost-number conservation. Free-field algebras with fermionic generators carry a $\mathbb{Z}$ -grading by ghost number preserved by the quadratic OPE. A mixed-weight insertion at a genus-0 vertex forces a net ghost-number mismatch that the quadratic vertex cannot absorb.

The triple redundancy is the structural reason free fields are the unique multi-weight survivors of the scalar formula. $\square$

Remark 2.3 (The propagator ratio). For a rank-2 algebra with curvatures κ_1, κ_2 and Zamolodchikov metric entries η_1, η_2 , the *propagator ratio*

$$\varrho_{12} := \frac{\eta^{22}}{\eta^{11}} = \frac{\eta_1}{\eta_2}$$

controls the strength of the cross-channel correction. When $\varrho_{12} = 1$, the system is effectively single-channel and $\delta F_g^{\text{cross}} = 0$. When $\varrho_{12} \neq 1$, the asymmetry drives nonzero cross-channel amplitudes. For $\mathcal{W}_3$: $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$, so $\varrho_{TW} = 3/2 \neq 1$.

3 $\mathcal{W}_3$ at genus 2: the seven stable graphs

3.1 The stable graph catalogue

The moduli space $\overline{\mathcal{M}}_{2,0}$ has seven stable graph types. The following table records each graph together with its edge count, automorphism group order, number of channel assignments for a rank-2 algebra, and cross-channel status.

Label	Name	$ E $	$ \text{Aut} $	Assign.	Cross status
A	smooth	0	1	1	none (no edges)
B	figure-eight	1	2	2	pure only
C	separating	1	2	2	genus-1 universal
D	sunset	2	8	4	mixed $\neq 0$
E	bridge-loop	2	2	4	mixed $\neq 0$
F	theta	3	12	8	mixed $\neq 0$
G	dumbbell	3	8	8	genus-1 universal

Graphs A, B, C, G contribute zero cross-channel correction:

- Graph A (smooth irreducible genus-2) has no edges.
- Graph B (figure-eight: one vertex of genus 0, one self-loop) has a single edge; only pure assignments exist.
- Graph C (separating: two genus-1 vertices joined by a bridge) has one edge. Each genus-1 factor is universal by Theorem 2.2(iii), so the clutching product is a product of genus-1-universal quantities.
- Graph G (dumbbell: two self-loops connected by a bridge, each self-loop a single-edge genus-1 component) has the same genus-1 universal structure.

Graphs D, E, F carry the entire cross-channel correction. Each has multiple edges incident to a common genus-0 vertex, so mixed-channel assignments produce genuine multi-channel vertex factors.

3.2 The computation

Computation 3.1 ($\mathcal{W}_3$ genus-2 cross-channel). The $\mathcal{W}_3$ algebra has rank 2 with channels T (weight 2, $\eta_{TT} = c/2$) and W (weight 3, $\eta_{WW} = c/3$). The inverse Zamolodchikov metrics are $\eta^{TT} = 2/c$ and $\eta^{WW} = 3/c$; the per-channel modular characteristics are $\kappa_T = c/2$ and $\kappa_W = c/3$, with total $\kappa(\mathcal{W}_3) = 5c/6$. The WW OPE contains the composite quasi-primary $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ with structure constant $16/(22+5c)$.

The three contributing graphs give:

Sunset (graph D). Two self-loops at a single genus-0 vertex with $n = 4$. The mixed assignments (T, W) and (W, T) give sunset amplitude $\delta F_2^{\text{sun}} = 3/c + 9/(2c) = 15/(2c)$ from the quartic vertex values and propagator products $\eta^{TT}\eta^{WW} = 6/c^2$, divided by $|\text{Aut}| = 8$.

Theta (graph F). Three edges between two genus-0 trivalent vertices. The mixed-channel assignments involve the coupling $C_{TWW} = c$ (the gravitational structure constant from the WWT OPE); odd-parity couplings $C_{TTW} = C_{WWW} = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$. The theta amplitude with mixed channels gives $\delta F_2^\theta = 1/16$.

Bridge-loop (graph E). One genus-1 vertex with a self-loop, connected by a bridge to a genus-0 trivalent vertex. The genus-1 vertex contributes $\kappa_j/24$, and the trivalent coupling C_{ijj} with propagators $\eta^{ii}\eta^{jj}$ gives $\delta F_2^{\text{bl}} = 21/(4c)$.

Summing:

$$\delta F_2^{\text{cross}}(\mathcal{W}_3) = \frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{c + 204}{16c} \quad (\text{ALL-WEIGHT}).$$

The full genus-2 free energy is therefore

$$F_2(\mathcal{W}_3) = \underbrace{\frac{7c}{6912}}_{\kappa \cdot \lambda_2^{\text{FP}} (\text{UNIFORM-WEIGHT})} + \underbrace{\frac{c + 204}{16c}}_{\partial F_2^{\text{cross}} (\text{ALL-WEIGHT})}.$$

Proposition 3.2 (Explicit $\partial F_2^{\text{cross}}(\mathcal{W}_3)$; wave-14 anchor). *For the $\mathcal{W}_3$ algebra at genus 2,*

$$\partial F_2^{\text{cross}}(\mathcal{W}_3) = \frac{c + 204}{16c} \quad (\text{ALL-WEIGHT}),$$

computed as a graph sum over the seven stable graphs of $\overline{\mathcal{M}}_{2,0}$ with rank-2 channel assignments. Only the sunset (D), theta (F), and bridge-loop (E) graphs contribute nonzero cross-channel amplitude; the remaining four graphs vanish by the genus-1 universality of the scalar formula (graphs C and G) or by edge count (graphs A and B). This proposition anchors the wave-14 scope of Theorem C(2) in Vol I: the multi-weight correction is explicit at $(\mathcal{W}_3, g = 2)$ and is the first instance where $\kappa \cdot \lambda_g^{\text{FP}}$ fails as the complete genus- g free energy for a multi-generator chirally Koszul algebra.

Proof. The sunset, theta, and bridge-loop amplitudes are computed in Computation 3.1. Summing:

$$\frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{120}{16c} + \frac{c}{16c} + \frac{84}{16c} = \frac{c + 204}{16c}. \quad \square$$

Remark 3.3 (Why the scalar formula fails). The cross-channel correction $\partial F_2^{\text{cross}} = (c + 204)/(16c)$ is nonzero for all $c > 0$. At large c , $\partial F_2^{\text{cross}} \rightarrow 1/16$, while the scalar part $\kappa \cdot \lambda_2^{\text{FP}} = 7c/6912 \rightarrow \infty$. The cross-channel correction is subleading at large c and genus 2, but this hierarchy reverses at higher genus. At $c = 204$, the cross-channel correction $\partial F_2 = 408/(16 \cdot 204) = 1/8$ is comparable to the scalar part $7 \cdot 204/6912 \approx 0.207$. The correction is a structural feature of multi-weight algebras, not a perturbative artifact.

4 $\mathcal{W}_3$ at genus 3: forty-two stable graphs

4.1 The genus-3 stable graph census

The moduli space $\overline{\mathcal{M}}_{3,0}$ has 42 stable graph types, stratified by vertex count: 1 one-vertex graph, 9 two-vertex, 14 three-vertex, 11 four-vertex, and 7 five-vertex. For the rank-2 algebra $\mathcal{W}_3$, each graph Γ admits $2^{|E(\Gamma)|}$ channel assignments.

Of the 42 graphs, exactly 34 contribute nonzero cross-channel corrections. The remaining 8 have cross-channel amplitude zero by channel-purity of single-edge genus-1 components: the same mechanism that kills graphs B, C, G at genus 2 (single-edge bridges connecting genus-1 subgraphs admit only pure channel assignments).

4.2 The genus-3 computation

Computation 4.1 ($\mathcal{W}_3$ genus-3 cross-channel). The genus-3 cross-channel correction for $\mathcal{W}_3$ is computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$, with all $2^{|E(\Gamma)|}$ channel assignments per graph. The

result:

$$\partial F_3^{\text{cross}}(\mathcal{W}_3) = \frac{5c^3 + 3792c^2 + 1,149,120c + 217,071,360}{138,240c^2} \quad (\text{ALL-WEIGHT}).$$

All four numerator coefficients are positive, so $\partial F_3^{\text{cross}}(\mathcal{W}_3) > 0$ for all $c > 0$.

Proof. Enumerating the 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ with all channel assignments gives the full genus-3 graph sum. The channel-purity mechanism removes 8 graphs whose single-edge genus-1 components force the mixed channel to vanish, leaving 34 contributing graphs. Evaluating the graph sum at enough integer values of c determines a unique rational function with denominator $138,240c^2$ and cubic numerator. Positivity follows from the positivity of every numerator coefficient. $\square$

4.3 Cross-channel dominance at genus 3

The large- c asymptotic of $\partial F_3^{\text{cross}}$ is

$$\partial F_3^{\text{cross}}(\mathcal{W}_3) \sim \frac{c}{27,648} \quad (c \rightarrow \infty),$$

which is *linear* in c , in contrast to the genus-2 correction which approaches the constant $1/16$. The net c -degree of $\partial F_g^{\text{cross}}$ increases with genus: degree 0 at $g = 2$, degree 1 at $g = 3$.

The ratio to the scalar part reveals a dramatic shift:

$$\frac{\partial F_3^{\text{cross}}(\mathcal{W}_3)}{\kappa(\mathcal{W}_3) \cdot \lambda_3^{\text{FP}}} \xrightarrow{c \rightarrow \infty} \frac{42}{31} \approx 1.35.$$

At genus 3 the cross-channel correction is already 35% larger than the scalar part in the large- c regime. For comparison, at genus 2 the ratio $\partial F_2^{\text{cross}}/(\kappa \cdot \lambda_2^{\text{FP}}) \rightarrow 0$ as $c \rightarrow \infty$: the scalar part dominates. The crossover occurs between genus 2 and genus 3.

Proposition 4.2 (Cross-channel dominance). *The full genus- g free energy of $\mathcal{W}_3$ satisfies (ALL-WEIGHT):*

$$F_g(\mathcal{W}_3) = \kappa(\mathcal{W}_3) \cdot \lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}(\mathcal{W}_3).$$

The ratio $\partial F_g^{\text{cross}}/(\kappa \cdot \lambda_g^{\text{FP}})$ grows at least factorially with genus, because $\kappa \cdot \lambda_g^{\text{FP}} = O(c \cdot (2\pi)^{-2g})$ decays exponentially in g while $\partial F_g^{\text{cross}}$ is bounded below by a polynomial in c whose degree grows with g . At sufficiently high genus, the cross-channel correction is the dominant contribution to the free energy.

Proof. By the Faber–Pandharipande formula, $\lambda_g^{\text{FP}} = (2^{2g-1}-1)|B_{2g}|/(2^{2g-1}(2g)!) = O(1/(2\pi)^{2g})$ (Bernoulli asymptotics), so $\kappa \cdot \lambda_g^{\text{FP}}$ decays exponentially in g . The cross-channel correction sums over $O(c^g)$ graphs (the number of stable graphs of $\overline{\mathcal{M}}_{g,0}$ grows exponentially), with graph amplitudes bounded polynomially in c . The resulting growth rate exceeds $\kappa \cdot \lambda_g^{\text{FP}}$ factorially. $\square$

Remark 4.3 (The genus tower). The explicit data through genus 4:

g	Formula for $\partial F_g^{\text{cross}}(\mathcal{W}_3)$ (ALL-WEIGHT)	Large- c	Graphs	$\lim_{c \rightarrow \infty} \frac{\partial F_g^{\text{cross}}}{\kappa \cdot \lambda_g^{\text{FP}}}$
2	$\frac{c + 204}{16 c}$	$\frac{1}{16}$	3 of 7	0
3	$\frac{5c^3 + 3792 c^2 + 1149120 c + 217071360}{138240 c^2}$	$\frac{c}{27648}$	34 of 42	$\frac{42}{31} \approx 1.35$
4	$\frac{287 c^4 + 268881 c^3 + \dots}{17418240 c^3}$	$\frac{41 c}{2488320}$	379 total	$\frac{9184}{381} \approx 24.1$

The pattern: the large- c limit of the ratio $\partial F_g^{\text{cross}}/(\kappa \cdot \lambda_g^{\text{FP}})$ is 0, 42/31, 9184/381 at genera 2, 3, 4. The growth is super-exponential in g ; by genus 4 the cross-channel correction exceeds the scalar part by a factor of 24.

5 The instanton hierarchy

The scalar genus expansion $F_g^{\text{scal}} = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) has a completely explicit resurgent structure, determined by the closed form

$$\sum_{g \geq 1} F_g^{\text{scal}} \hbar^{2g} = \kappa \left(\frac{\hbar/2}{\sin(\hbar/2)} - 1 \right).$$

The poles at $\hbar = 2\pi n$ for $n \geq 1$ produce Borel singularities at $\xi_n = (2\pi n)^2$, and the leading instanton action is $A_{\text{scal}} = (2\pi)^2$.

Proposition 5.1 (Scalar Borel transform). *The Borel transform of the scalar genus series in the variable $u = \hbar^2$,*

$$\mathcal{B}[Z](\xi) := \sum_{g \geq 1} \frac{F_g^{\text{scal}}}{(g-1)!} \xi^{g-1},$$

is an entire function of ξ (UNIFORM-WEIGHT). The Borel coefficients satisfy

$$\frac{|F_g^{\text{scal}}|}{(g-1)!} \sim \frac{2|\kappa|}{(2\pi)^{2g} (g-1)!},$$

which decays superexponentially in g . The radius of convergence is infinite.

Proof. The Bernoulli asymptotics $|B_{2g}| \sim 2(2g)!/(2\pi)^{2g}$ give $|F_g^{\text{scal}}| = O(1/(2\pi)^{2g})$. Dividing by $(g-1)!$ produces superexponential decay. The root test gives infinite radius. $\square$

Proposition 5.2 (Universal instanton action and trans-series). *Under the uniform-weight hypothesis, the shadow partition function has the trans-series form (UNIFORM-WEIGHT):*

$$Z^{\text{full}}(\hbar) = \sum_{n \geq 0} \sigma^n e^{-n \cdot A_{\text{scal}}/\hbar^2} Z^{(n)}(\hbar),$$

where $A_{\text{scal}} = (2\pi)^2$ is the universal instanton action, σ is the trans-series parameter, and $Z^{(0)} = Z_{\text{scal}}^{\text{sh}}$ is the perturbative sector. The leading Stokes multiplier is $\mathfrak{S}_1 = -4\pi^2 \kappa(\mathcal{A}) i$.

For multi-weight algebras $(\mathcal{W}_3, \mathcal{W}_4, \dots)$: the scalar instanton action $A_{\text{scal}} = (2\pi)^2$ still governs the scalar free energy $F_g^{\text{diag}} = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT), but the cross-channel correction $\delta F_g^{\text{cross}}$ (ALL-WEIGHT) carries its own large-order asymptotics that are not controlled by a single universal instanton action. The polynomial degree of $\delta F_g^{\text{cross}}$ in c grows with genus, so the cross-channel asymptotics exhibit c -dependent growth. For $\mathcal{W}_3$: the ratio $\delta F_3^{\text{cross}} / \delta F_2^{\text{cross}} \sim c/1728$ at large c , growing without bound. The cross-channel genus expansion grows faster than the scalar expansion at large c .

Proof. The trans-series structure follows from the closed-form expression for the scalar partition function, whose poles at $\hbar = 2\pi n$ produce the Borel singularities. The instanton action $A_{\text{scal}} = \xi_1 = (2\pi)^2$ depends only on the $\hat{A}$ -genus normalization, not on the algebra. The Stokes multiplier $\mathfrak{S}_n = (-1)^n \cdot 4\pi^2 n \cdot \kappa \cdot i$ is computed from the residues of the closed form.

For the cross-channel part, the explicit data gives: at genus 2, $\delta F_2^{\text{cross}}(\mathcal{W}_3) = O(1)$ at large c ; at genus 3, $\delta F_3^{\text{cross}}(\mathcal{W}_3) = O(c)$. The ratio $\delta F_3^{\text{cross}} / \delta F_2^{\text{cross}}$ is a rational function of c with positive leading term:

$$\frac{\delta F_3^{\text{cross}}}{\delta F_2^{\text{cross}}} \sim \frac{c/27648}{1/16} = \frac{c}{1728} \quad (c \rightarrow \infty).$$

This unbounded growth rules out a single cross-channel instanton action in the usual Borel sense. $\square$

Remark 5.3 (Gevrey classification). The scalar genus expansion is Gevrey-1: the ratio $F_{g+1}^{\text{scal}} / F_g^{\text{scal}} \sim (2g+1)(2g+2)/(2\pi)^2$ grows quadratically in g , confirming factorial divergence with instanton action $A_{\text{scal}} = (2\pi)^2$ (UNIFORM-WEIGHT). This is in sharp contrast with string amplitudes, where the ratio grows as $(2g)!$ and the Borel transform has finite radius. The shadow expansion has an *entire* Borel transform because the Bernoulli decay $1/(2\pi)^{2g}$ kills the factorial growth.

The cross-channel sector has a qualitatively different Gevrey structure. The polynomial-degree growth of $\delta F_g^{\text{cross}}$ in c means that at fixed c the cross-channel series is factorially divergent (Gevrey-1), but at large c the effective instanton action $A_{\text{cross}}(c)$ grows with c (ALL-WEIGHT). The hierarchy $A_{\text{cross}}(c) > A_{\text{scal}} = (2\pi)^2$ at large c is the quantitative statement that cross-channel saddles are heavier than scalar saddles.

6 Universal N -formula for $\delta F_2(\mathcal{W}_N)$

6.1 The gravitational Frobenius algebra

Definition 6.1 (Gravitational Frobenius algebra of $\mathcal{W}_N$). The principal $\mathcal{W}$ -algebra $\mathcal{W}_N$ has strong generators $W^{(2)}, W^{(3)}, \dots, W^{(N)}$ at conformal weights $2, 3, \dots, N$ ($N-1$ generators; highest weight N). The *gravitational Frobenius algebra* of $\mathcal{W}_N$ is the rank- $(N-1)$ Frobenius algebra with per-channel inverse Zamolodchikov metric $\eta^{(j)(j)} = j/c$ for $j = 2, \dots, N$ (from the leading-pole normalization $\eta_{(j)(j)} = c/j$) and structure constants $C_{(i)(j)(k)}^{\text{grav}} = c \cdot \delta_{ijk}^{\text{even}}$, nonzero only when the number of odd-weight indices is even (reflecting the $\mathbb{Z}_2$ parity of odd-spin generators).

6.2 The closed-form formula

Proposition 6.2 (Universal gravitational cross-channel formula for $\mathcal{W}_N$). (i) Closed-form formula. *The genus-2 gravitational cross-channel correction is (ALL-WEIGHT):*

$$\partial F_2^{\text{grav}}(\mathcal{W}_N, c) = \frac{(N-2)(N+3)}{96} + \frac{(N-2)(3N^3 + 14N^2 + 22N + 33)}{24c}.$$

Equivalently,

$$\partial F_2^{\text{grav}}(\mathcal{W}_N, c) = \frac{(N-2)[c(N+3) + 4(3N^3 + 14N^2 + 22N + 33)]}{96c}.$$

- (ii) Vanishing characterization. $\partial F_2^{\text{grav}}(\mathcal{W}_N, c) = 0$ for all $c > 0$ if and only if $N = 2$ (Virasoro: uniform-weight, no cross-channel correction).
- (iii) Strict positivity. For $N \geq 3$ and all $c > 0$, $\partial F_2^{\text{grav}}(\mathcal{W}_N, c) > 0$.
- (iv) Large- c and large- N asymptotics. Define the leading coefficient $B(N) := (N-2)(N+3)/96$ and the subleading coefficient $A(N) := (N-2)(3N^3 + 14N^2 + 22N + 33)/24$, so that $\partial F_2^{\text{grav}} = B(N) + A(N)/c$ (ALL-WEIGHT). At large N :

$$B(N) \sim \frac{N^2}{96}, \quad A(N) \sim \frac{N^4}{8}.$$

- (v) Explicit specializations.

$$\mathcal{W}_3: \quad \partial F_2^{\text{grav}} = \frac{c + 204}{16c} \quad (\text{exact}),$$

$$\mathcal{W}_4: \quad \partial F_2^{\text{grav}} = \frac{7c + 2148}{48c} \quad (\text{lowerbound}),$$

$$\mathcal{W}_5: \quad \partial F_2^{\text{grav}} = \frac{c + 434}{4c} \quad (\text{lowerbound}).$$

- (vi) Exactness and lower-bound property. For $\mathcal{W}_3$, the gravitational formula is **exact**: the $\mathbb{Z}_2$ parity $W \mapsto -W$ kills all higher-spin exchange couplings ($C_{TTW} = C_{WWW} = 0$). For $N \geq 4$, higher-spin exchange structure constants are generically nonzero and positive, so the full cross-channel correction satisfies

$$\partial F_2(\mathcal{W}_N, c) \geq \partial F_2^{\text{grav}}(\mathcal{W}_N, c) \quad (\text{ALL-WEIGHT})$$

with equality if and only if $N \leq 3$.

Proof. The genus-2 cross-channel correction is computed by summing over the three contributing stable graphs D (sunset), E (bridge-loop), F (theta) from the catalogue of §3, with all mixed-channel assignments $\sigma: E(\Gamma) \rightarrow \{2, \dots, N\}$.

Sunset (graph D). The quartic vertex value $V_{o,4}(i, i, j, j) = \sum_k C_{iik} \eta^{kk} C_{jjk}$ depends on i and j only through parity. For even-weight channels: $V_{o,4} = 2c$. The sunset cross-channel contribution involves mixed assignments with propagator products $\eta^{(i)(i)} \eta^{(j)(j)} = ij/c^2$, divided by $|\text{Aut}| = 8$.

Theta (graph F). The trivalent vertex $C_{(i)(j)(k)}$ is nonzero only for even-parity triples. The theta amplitude $C_{ijk}^2 \eta^{ii} \eta^{jj} \eta^{kk} / 12$ sums over all nonzero even-parity triples.

Bridge-loop (graph E). The genus-1 vertex contributes $\kappa_j / 24$ (genus-1 universality), and the genus-0 trivalent vertex contributes C_{ijj} . The amplitude is $C_{ijj} \eta^{ii} \eta^{jj} \kappa_j / 48$.

The closed-form coefficients $B(N) = (N-2)(N+3)/96$ and $A(N) = (N-2)(3N^3 + 14N^2 + 22N + 33)/24$ are obtained by Newton interpolation from the graph sum evaluated at $N = 3, 4, \dots, 11$ and verified at 5 or more values of c per N . The common factor $(N-2)$ reflects the vanishing at $N = 2$ (Virasoro: unique conformal weight).

Part (iii): $B(N) > 0$ and $A(N) > 0$ for $N \geq 3$ because $(N-2) > 0$, $(N+3) > 0$, and $3N^3 + 14N^2 + 22N + 33 > 0$ for all $N \geq 3$.

Part (vi): for $\mathcal{W}_3$, the only even-parity triples involving the W -channel are (T, W, W) and permutations. The cross-coupling $C_{TWW}^{\text{grav}} = c$ accounts for all gravitational exchange; $C_{TTW} = C_{WWW} = 0$ eliminates all higher-spin exchange. For $N \geq 4$, even-parity triples $(W^{(a)}, W^{(b)}, W^{(c)})$ with $a, b, c \neq 2$ have generically nonzero positive structure constants. $\square$

Remark 6.3 (N -degree pattern). The polynomial degree in N of the coefficient of c^{-j} in $\partial F_g^{\text{grav}}(\mathcal{W}_N)$ satisfies

$$\deg_N(e_{g,-j}(N)) = 2j + g \quad (j = 0, 1, \dots, g-1).$$

At genus 2: $e_{2,-1}(N) = A(N)/D_2$ has $\deg_N A = 4 = 2 \cdot 1 + 2$; $e_{2,0}(N) = B(N)/D_2$ has $\deg_N B = 2 = 2 \cdot 0 + 2$. The universal vanishing factor $(N-2)$ accounts for 1 of the N -degree; the residual degree is $2j + g - 1$. This pattern extends to genus 3, where $\deg_N A_3 = 7 = 2 \cdot 2 + 3$, $\deg_N B_3 = 5 = 2 \cdot 1 + 3$, $\deg_N C_3 = 3 = 2 \cdot 0 + 3$.

Remark 6.4 (Spectral curve obstruction). No spectral curve with at most two branch points reproduces the cross-channel tower $\{\partial F_g^{\text{cross}}(\mathcal{W}_3)\}_{g \geq 2}$. The Newton identity test: define Bernoulli-normalized power sums $\sigma_g := \partial F_g / (B_{2g} / (2g(2g-2)))$. If ∂F_g arose from topological recursion on a spectral curve with ≤ 2 branch points, the σ_g would satisfy a linear recurrence of order ≤ 2 . The recurrence holds at $g = 2, 3$ but fails at $g = 4$ with relative error 18–27%. The correct framework is CohFT-weighted topological recursion on the $\mathcal{A}_2$ Frobenius manifold.

7 The E_1 interpretation: cross-channel dominance as E_1 primacy

7.1 The ordered bar resolves all channels

The primitive object of the modular Koszul programme is the ordered bar complex $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ with deconcatenation coproduct and differential from the E_1 -chiral product. The E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$ carries the classical r -matrix $r(z)$ as the genus-0 binary collision residue of the E_1 -framed MC element $\Theta_{\mathcal{A}}^{E_1}$. The symmetric bar $B^{\Sigma}(\mathcal{A})$ is the Σ_n -coinvariant shadow, and the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is the image of the lossy coinvariant projection $\text{av}: \mathfrak{g}_{\mathcal{A}}^{E_1} \rightarrow \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$.

The five theorems of the programme extract the Σ_n -invariant content of the ordered bar. The modular characteristic is $\kappa(\mathcal{A}) = \text{av}(r(z))$ at degree 2 in the abelian and scalar families; for non-abelian affine Kac-Moody, $\kappa(V_k(\mathfrak{g})) = \text{av}(r(z)) + \dim(\mathfrak{g})/2$. The scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) is the statement that the coinvariant projection loses no genus information for uniform-weight algebras.

7.2 The cross-channel correction as averaging loss

For multi-weight algebras, the coinvariant projection loses information. The ordered bar $B^{\text{ord}}(\mathcal{A})$ resolves all r weight channels simultaneously: each tensor factor in $T^c(s^{-1}\tilde{\mathcal{A}})$ carries a channel label, and the deconcatenation coproduct preserves channel assignments. The stable-graph expansion of the MC element $\Theta_{\mathcal{A}}^{E_1}$ computes the full genus- g free energy with all channel data intact: every edge of every stable graph carries a definite channel label, and the graph sum over $\overline{\mathcal{M}}_{g,o}$ includes all $r^{|E(\Gamma)|}$ channel assignments.

The coinvariant projection av collapses channel labels to their symmetric average. In the E_1 picture, the diagonal channel assignments (all edges in the same channel) are Σ_n -invariant and survive the projection; the mixed channel assignments break the permutation symmetry and are killed. The cross-channel correction $\delta F_g^{\text{cross}}$ is precisely the content of $\ker(\text{av})$ at genus g : it is the information about the ordered bar complex that the coinvariant projection discards.

Proposition 7.1 (E_1 characterization of the cross-channel correction). *The cross-channel correction decomposes as (ALL-WEIGHT):*

$$\delta F_g^{\text{cross}}(\mathcal{A}) = F_g^{E_1}(\mathcal{A}) - F_g^{\Sigma}(\mathcal{A}),$$

where $F_g^{E_1}(\mathcal{A})$ is the genus- g free energy computed from the ordered bar complex (with all channel assignments), and $F_g^{\Sigma}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ is the coinvariant projection (UNIFORM-WEIGHT scalar lane). The cross-channel correction vanishes if and only if the coinvariant projection is lossless at genus g .

7.3 Quantitative vindication of E_1 primacy

The data of §4 gives a quantitative measure of E_1 primacy. At genus 2, the E_1 information beyond the coinvariant projection contributes $(c+204)/(16c)$ to the free energy: a correction that is subleading at large c but nonzero at every finite c . At genus 3, the ordered-bar information already dominates: the ratio $\delta F_3^{\text{cross}}/F_3^{\Sigma} \rightarrow 42/31$ at large c means that more than half the genus-3 free energy lives in $\ker(\text{av})$, invisible to the symmetric bar. At genus 4, the ratio $9184/381 \approx 24$ means that 96% of the free energy is cross-channel: the scalar lane contributes only 4%.

This is the precise quantitative content of E_1 primacy at multi-weight. The ordered bar $B^{\text{ord}}(\mathcal{A})$ is not merely the technically convenient resolution of $B^{\Sigma}(\mathcal{A})$; it is the primary object that carries most of the genus- g information. The symmetric bar, and the scalar formula that comes with it, captures only the tip of the genus expansion. The cross-channel correction is the ordered bar's quantitative vindication: the deeper the genus, the more information the coinvariant projection destroys, and the larger the cross-channel correction relative to the scalar part.

Remark 7.2 (Irreducible bivariate). The generating function $\Delta(c, \hbar) := \sum_{g=2}^{\infty} \delta F_g^{\text{cross}}(\mathcal{W}_N, c) \hbar^{2g}$ (ALL-WEIGHT) admits no closed-form expression in one variable. For $\mathcal{W}_3$, it is an *irreducibly bivariate* function of $(c, \hbar)$: it does not factor as $f(\hbar) \cdot h(c)$, it is not generated by an $\hat{\mathcal{A}}$ -genus ansatz, and the numerator polynomial $P_g(c)$ in $\delta F_g = P_g(c)/(D_g c^{g-1})$ changes qualitative structure with each genus. Five independent obstructions establish this: inhomogeneous c -scaling (c -power window widens with genus), super-linear ratio growth (consecutive ratios are non-polynomial in c), non-separability (cross-ratio test fails), $\hat{\mathcal{A}}$ -genus obstruction ($\delta F_3/(\delta F_2)^2$ varies with c), and irreducible numerators (P_2, P_3, P_4 admit no uniform Euler-product factorization).

Remark 7.3 (Free fields as unique survivors). The free-field algebras (Heisenberg, $\beta\gamma$, bc , lattice) are the unique multi-weight algebras for which the coinvariant projection is lossless: $\partial F_g^{\text{cross}} = 0$ for all $g \geq 1$. In the E_1 picture, this is the statement that the ordered bar complex of a free-field algebra carries no more genus information than its coinvariant shadow. The triple redundancy of the proof (shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation) expresses three independent projections of the same geometric fact: the quadratic free-field OPE factorizes the propagator through weight sectors, and the ordered bar complex accordingly decomposes as a product of single-channel bars. Products have no cross-terms.

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